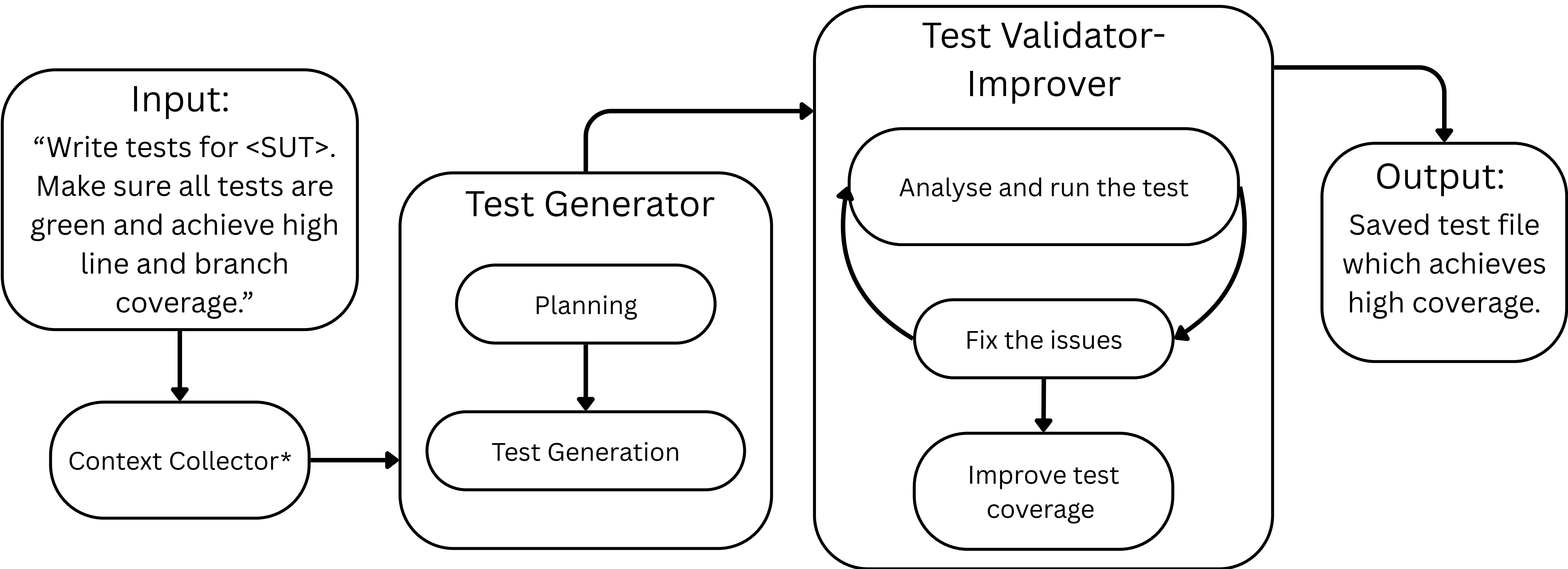


TESTGEN: Agentic Test Generation

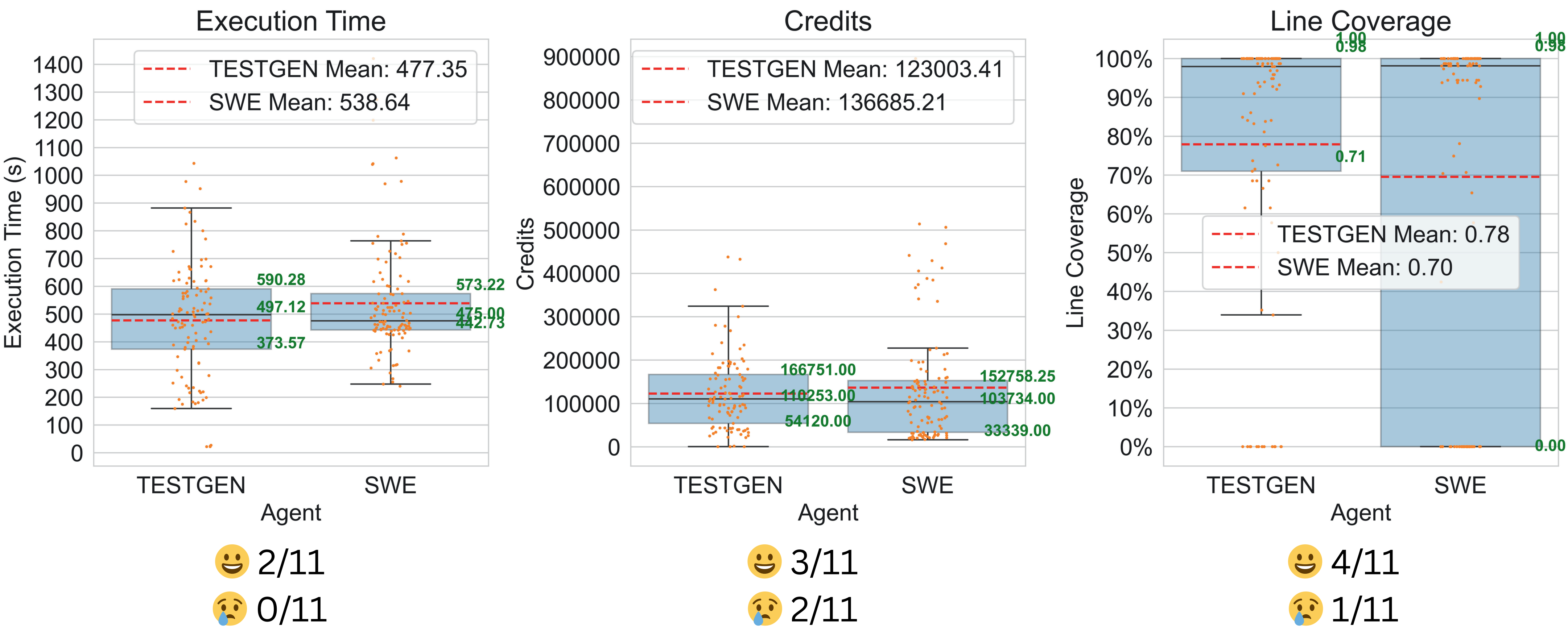
Software Testing Research Team

Agentic approach to test generation



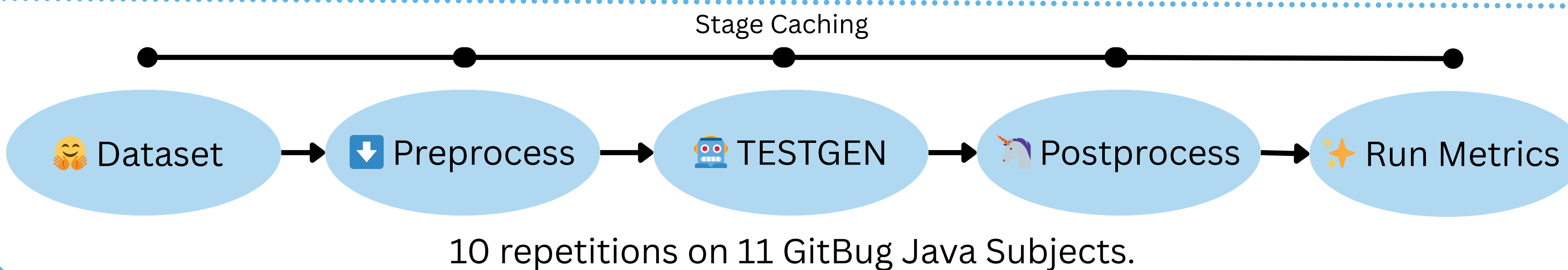
*We designed a custom context collector with emphasis on time and credit efficiency. For more details, please refer to the recording of our AR seminar. [📺](#)

Results



😊 - Statistically significant TESTGEN win | 😞 - Statistically significant TESTGEN loss

Evaluation



Lessons

1. Divide big tasks and conquer them.
2. No free lunch – customise your strategy.
3. Minimize agent’s decision making.
4. Be specific with enough context.

Future Steps

1. Support more languages.
2. Support more test generation scenarios (e.g., evaluate the agent against SWTBench)
3. Compare with Junie instead of SWE.

